MIL-S-19500/450A(EL)
10 August 1973
SUPERSEDING
MIL-S-19500/450(EL)
2 June 1971

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, POWER

TYPES 2N5958, 2N5960, TX2N5958, AND

TX2N5960

1. SCOPE

1.1 Scope – This specification covers the detail requirements for silicon, PNP, power transistors for particular use in power switching, electronic circuit applications. (See 3.3 and 6.2 herein.) The prefix "TX" identifies devices meeting the special process-conditioning, testing, and screening requirements in 4.4 herein, and the stringent quality-control (LTPD) requirements in Tables I, II, and III herein.

1.2 Outline and dimensions – See Figure 1 herein.

1.3 Maximum Ratings

	P_T 1/							
	at: +50°C	at: T_C = +100°C	Θ_{J-C}	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T(oper) T (stg)
	W	W	°C/W	Vdc	Vdc	Vdc	Adc	°C
2N5958	150	100	1	100	100	10	20	2/
2N5960	150	100	1	100	100	10	20	2/

1/ Linear derating factor = 1.0 W/°C up to and including T_C = +200°C.

2/ Temperature range = -65°C to +200°C.

FSC 5961

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1.4 Particular electrical characteristics – (At T_C = +25°C)

	h_{FE} 1/						/hfe/	
	at: I_C = 1 Adc, V_{CE} = 10 Vdc		at: I_C = 10 Adc, V_{CE} = 10 Vdc		at: I_C = 20 Adc, V_{CE} = 10 Vdc		at: f = 10 MHz, V_{CE} = 15 Vdc, I_C = 2 Adc	
	Min	Max	Min	Max	Min	Max	Min	Max
2N5958	40	–	30	120	10	–	1.0	8.0
2N5960	40	–	30	120	10	–	1.0	8.0

	$V_{CE(sat)}$ 1/				$V_{BE(sat)}$ 1/				t_{on}	t_{off}
	at: I_C = 10 Adc, I_B = 1 Adc		at: I_C = 20 Adc, I_B = 2 Adc		at: I_C = 10 Adc, I_B = 1 Adc		at: I_C = 20 Adc, I_B = 2.0 Adc		at: I_C = 20 Adc (See Figure 3 herein)	
	Min	Max	Min	Max	Min	Max	Min	Max	Max	Max
	Vdc	Vdc	Vdc	Vdc		Vdc		Vdc	usec	usec
2N5958	–	0.9	–	1.8	–	1.5	–	2.0	1.1	1.2
2N5960	–	0.9	–	1.8	–	1.5	–	2.0	1.1	1.2

1/ Pulsed: see 4.3.1 herein.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification for

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. — The following document forms a part of the specification to the extent specified herein. Unless otherwise noted, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Requirements. - Requirements for the transistors shall be in accordance with Specification MIL-S-19500, and as otherwise specified herein.

3.2 Design and construction. - The Transistor shall be of the design, construction, and physical dimensions specified in Figure 1.

3.2.1 Terminal arrangement. - The terminal arrangement on the transistors shall be as indicated in Figure 1.

3.2.2 Operating position. - The transistor shall be capable of proper operation in any position.

3.3 Performance characteristics. - Performance characteristics shall be as specified in Tables I, II and III and as follows:

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3.3.1 "TX" special requirements. - Specified "TX" device types shall, in addition to all other performance requirements herein, be process-conditioned, tested, and screened in accordance with 4.4 herein (see 4.3.1 herein.) (At option of the manufacturer, the non-TX device types may be subjected to the same process-conditioning and testing.)

3.4 Marking. - Marking shall be in accordance with Specification MIL-S-19500.

3.4.1 "TX" identification marking. - Devices in accordance with the "TX" requirements shall be marked with "TX" immediately following the "JAN" or "J" prefix.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and Quality conformance inspection. - Qualification and Quality Conformance Inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II, and III respectively, herein. Quality Conformance Inspection shall include inspection of Preparation for Delivery (see 5.1 herein.)

4.2.1 Permissible Group B or Group C combined-subgroup testing. - At option of the manufacturer, all of the tests of one subgroup in Group B (Table II herein) may be combined with all of the tests in one or more other subgroups of Group B for sequential performance; the same option applies to Group C (Table III herein) constituent subgroups. The sample size and accept-reject criteria for such combined subgroup testing shall be based upon the existing LTPD or lambda (λ) for each of the subgroups involved in the particular combination. The following administrative requirements shall apply hereto:

a) It shall be understood that the above option may be adopted by manufacturer for Qualification testing of product, or for Quality Conformance inspection o f any individual lot submitted by manufacturer for lot acceptance.

b) Prior notification as to adoption of this testing option, and concurrent information as to the predesignated sample quantities to be used shall be made to the Government inspection authority concerned.

c) Complete record of the combined subgroup testing program results shall be maintained and be available to the Government inspection authority.

4.2.2 Qualification testing. - The non-TX device types shall be considered subject to Qualification tests. It shall be understood that procurement of the TX device types will be made from sources for the approved non-TX devices as listed in Qualified Products List (QPL)-19500. Upon request to and validation by the qualifying activity, qualification may be extended (and appropriate QPL listing effected) to TX devices manufactured by approved sources for the non-TX devices.

4.2.2.1 Table 1 Acceptance Criteria for Qualification Inspection. - For qualification inspection, all units that are to be subjected to Group B and C inspection shall first be subjected to all tests of Group A inspection using a combined LTPD of 5 for Group A acceptance criteria.

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4.2.3 Group B and C life test samples. - Samples that have been subjected to Group B, 340 hour life test, may be continued on test to 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1000 hour acceptance evaluation after they have passed the Group B 340 hour acceptance criteria. The cumulative total of failures found during 340 hour test and during the subsequent interval up to 1000 hours shall be computed for 1000 hour acceptance criteria.

4.2.4 Disposition of sample units. - Sample units that have been subjected to Group B, subgroup 2 and 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroup 1, 3, 4, 6, 7, 8, 9, and 10 and all Group C tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed Group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.2.5 Copies of lot inspection data. - When specified in the contact or order, one copy of the Quality Conformance inspection data, including manufacturer's test data and findings, covering the device inspection lot being shipped shall accompany the device(s) concerned.

4.3 Particular examination and test requirements. -

4.3.1 "Pulsed" - conditions measurements. - Measurements required herein to be effected under pulsed conditions, shall be made in accordance with "Pulse Measurements" requirements in Section 4 of Standard MIL-STD-750.

4.3.2 Mechanical damage resulting from tests. - Except for intentionally deforming, mutilating, or dismembering mechanical stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Groups A, B, or C Tests.

4.3.2.1 Durability of terminal lugs. - The lug ends of terminals shall remain intact on terminals throughout all physical, mechanical, and environmental tests herein.

4.3.3 Terminal Strength (Stud Torque) test. - Acceptance criteria for external threaded parts (see Fig. 1 herein) shall be in accordance with Handbook H-28.

4.3.4 Safe Operating Area (DC Operation) test. - Each transistor under test shall sustain the applicable test conditions and the following acceptance criteria shall apply:

a) I_C (for each transistor) shall not vary more than + 10% during the dc operation, and

b) All other specified End-Point Test(s) limits shall not be exceeded after the dc operation test.

c) Correlation note: Satisfactory endurance of the transistors throughout tests 1, 2, 3, respectively (Table II, Subgroup 6) is directly associable with ascertainment of the "safe operating area" for the transistors as illustrated in the nomograph of Figure 4 herein.

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4.3.5 Marking Resistance to Solvents test. - The device samples shall be subjected to test per Method 215 in standard MIL-STD-202, except that the following details shall apply:

a) All surface areas on the body of the device where marking has been applied shall be brushed.

b) All marking shall have remained legible, and there shall be no evidence of mechanical damage to the device, upon examination after test.

4.3.6 Operating life test condition. Power shall be applied to the device to achieve a junction temperature(T_J) of +200 °C ± 12.5°C.

$T_J = P_D \theta_{JC} + T_C$; $P_D = (V_{CE})(I_C)$; $\theta_{JC} = 1.0°$ C/W

T_C = measured value at hottest point on case.

Note: For reference only - This test can be approximated by dissipating 6 watts at T_A = + 25°C or 106 watts at T_C = + 100°C.

4.3.7 Group C life test. The steady-state operation life test (method 1027) in subgroup 4 of group C need not be conducted as a group C test if performed in subgroup 10 of group B as an alternative to the T_J steady-state operation life test. If the option of T_C testing is conducted in subgroup 10 of group B the same test sample maybe continued in subgroup 3 of group C.

4.4 Process conditioning, testing, and screening for "TX" devices. - All "TX" devices (100% of each lot) shall be subjected to the process conditioning, testing, and screening specified in 4.4.2 through 4.4.7.4 herein, in accordance with the procedural sequence established in Figure 2 herein, prior to submission of the lot to Group A, B, and C tests. The manufacturer shall permit the authorized Government representative to witness concurrently with time of manufacturers performance of these tests, the process conditioning, testing, and screening on the devices. Those conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than those specified herein.

4.4.1 Quality Assurance (lot verification.) – The manufacturer shall maintain and make available for perusal and review at any time, during a period of 1 year minimum, to the authorized Government representative or other contracting customer concerned, complete lot by lot records for these devices. All devices that fail to meet the process conditioning, testing, and screening requirements specified herein shall be removed from the respective lot, and the test failure identity and quantity removed shall be recorded in the permanent lot history. (See 6.2, 6.3 and 6.4 herein.)

4.4.2 High temperature storage. – The transistors shall be stored for at least 24 hours at a minimum temperature (T_A) of +200°C.

4.4.3 Thermal shock (temperature cycling). – The transistors shall be subjected to temperature cycling (thermal shock) in accordance with MIL-STD-750, Method 1051, Test Condition C, except that the cycling duration shall be 10 continuous cycles and exposure time at temperature extremes shall be 15 minutes minimum.

4.4.4 Acceleration. – The transistors shall be subjected to the constant acceleration (centrifugal) per Method 2006 in Standard MIL-STD-750 except that the following shall apply; a) centrifugal force = 5000G; b) Orientation = Y1, only; and c) hold time of 1 minute shall not apply hereto.

4.4.5 Hermetic seal. - The transistors shall be subjected to hermetic seal tests in accordance with requirements of Method 1071 in Standard MIL-STD-750 except that the following test conditions therein shall apply hereto:

a) Fine leak test: Per Test Condition G or H, however, a maximum leakage rate of 5×10^{-7} atm cc/sec shall be effective for acceptance.

b) Gross leak test: per Test Condition A, C, D, or F.

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4.4.6 Reverse bias burn-in – The transistors shall be operated for 48 + 4 hours minimum, at T_A = +150°C, under the following test conditions:

$$V_{CB} = 80 \text{ Vdc}, I_E = 0$$

At end of the 48 hour period, the above specified test voltages shall be maintained on the transistors until T_A = +30°±5°C has been reached. Within 4 hours after removal of the bias voltage, and while the transistors have been maintained at the specified room ambient temperature without any other voltages applied, they shall be subjected to measurement for I_{CES} in accordance with Table IV below.

4.4.7 Burn-in conditioning, tests, and evaluation (screening). – To enable proper evaluation of the transistors conditioned and tested in accordance with the following requirements, each transistor and lot shall be handled or identified in such manner that results for each individual transistor and lot can be ascertained and recorded.

4.4.7.1 Pre burn-in measurements. – The transistors shall be tested for the characteristics listed in Table IV below, in the test sequence indicated, and the values found shall be recorded. Those transistors that fail to meet the specified limits shall be rejected and removed from the lot.

TABLE IV. Burn-in tests measurements.

Test Method per MIL-STD-750	Examination or Test	Test Conditions	Symbol	Limits Min	Limits Max	Unit
3041	Collector to emitter cutoff current	Bias Cond. C V_{CE} = 100 Vdc	I_{CES}	---	100	uAdc
3076	Forward current transfer ratio	Pulsed; see 4.3.1 V_{CE} = 10 Vdc I_C = 10 Adc	h_{FE}	30	120	---

4.4.7.2 Burn-in conditioning. - The transistors shall be operated under the following conditions for 168+4 hours:

T_J = +200°C ± 12.5°C
V_{CE} = 10 to 25 volts
(see 4.3.6)

4.4.7.3 Post burn-in tests and measurements. - After burn-in conditioning, the transistors shall be permitted to return to room ambient temperature thermal equilibrium and re-tested, within a maximum total elapsed time of 96 hours, for the characteristics listed in Table IV above. The values now found for each individual transistor shall not exceed the following (respectively applicable to the pre burn-in measurements per Table IV above):

I_{CES} = ±100%, or 10 uA, whichever is greater.
h_{FE} = ±20%.

4.4.7.4 Burn-in conditioning evaluation (screening). - Those transistors whose measured characteristics exceed the delta limits specified in 4.4.7.3 above, or the limits of Table IV, shall be rejected and removed from the lot. Where 10% or more of the transistors subjected to the post burn-in tests are found to exceed those limits of Table IV, the entire inspection lot shall be considered non-acceptable as "TX" devices.

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4.5 Unclamped Inductive Sweep test. - The following particular test and circuit-element details, additional to those specified in Table II, Subgroup 7 herein for this test, shall apply:

Duty cycle ≤ 10%	L = 5 mH+5%; R = 0.03 ohms, I = 22.5 amps
$t_p \simeq 375$ usec	(Stancor C2689 or equivalent)
$V_{BB1} = 12$ Vdc	$R_L = 0$
$V_{BB2} = 0$	$R_S = 1$ ohm, 12W, 1% tolerance
$R_{BB1} = 25$ ohms	non-inductive
$R_{BB2} = 00$	VCC = 40V
	$I_C = 3A$

4.6 Clamped Inductive Sweep test. - The following particular test and circuit-element details, additional to those specified in Table II, Subgroup 8 herein for this test, shall apply:

Clamp voltage = 100V	L = 5 mH+5%; R = 0.03 ohms, I = 22.5 amps
A suitable clamping circuit or a	(Stancor C2689 or equivalent)
clamping diode may be used.	$R_L = 0$
Duty cycle ≤ 10%	$R_S = 1$ ohms; 12W; 1% tolerance,
$t_p \simeq 1$ msec	(non-inductive)
$V_{BB1} = 12$ V	$R_{BB1} = 5$ ohms
$V_{BB2} = 1.5$V	$R_{BB2} = 100$ ohms
$V_{CC} = 100$V	$I_C = 20A$

Note: device fails if clamp voltage is not reached.

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Table 1. — Group A inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
2071	Subgroup 1 Visual and mechanical examination	---	10	5	---	---	---	---
	Subgroup 2							
3041	collector-emitter cutoff current	Bias Cond. C V_{BE} = 0 V_{CE} = 100 Vdc	5	3	I_{CES}	---	100	uAdc
3061	Emitter-base cutoff current	Bias Cond. D V_{EB} = 8 Vdc			I_{EBO}	---	100	uAdc
3001	Collector-base breakdown voltage	Bias Cond. D I_C = 500 uAdc Pulsed; (see 4.3.1)			BV_{CBO}	100	---	Vdc
3011	Collector-emitter breakdown voltage	Bias Cond. C I_C = 200 mAdc Pulsed; (see 4.3.1)			BV_{CEO}	100	---	Vdc
3026	Emitter-base breakdown voltage	Bias Cond. D I_E = 500 uAdc			BV_{EBO}	10	---	Vdc
	Subgroup 3							
3076	Forward current transfer ratio	Pulsed; (see 4.3.1) I_C = 1 Adc V_{CE} = 10 Vdc	5	3	h_{FE}	40	---	---
3076	Forward current transfer ratio	Pulsed; (see 4.3.1) I_C = 10 Adc V_{CE} = 10 Vdc			h_{FE}	30	120	---
3076	Forward current transfer ratio	Pulsed; (see 4.3.1) I_C =20 Adc V_{CE} = 10 Vdc			h_{FE}	10	---	---
3066	Base-emitter saturation voltage	Test Cond. A I_C = 10 Adc I_B = 1 Adc Pulsed; (see 4.3.1)			VBE(sat)	---	1.5	Vdc

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Table 1. – Group A inspection (cont'd)

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 3 (cont'd)							
3066	Base-emitter saturation voltage	Test Cond. A I_C = 20 Adc I_B = 2 Adc Pulsed; see 4.3.1			V_{BE}(sat)	---	2.0	Vdc
3071	Collector-emitter saturation voltage	I_C = 10 Adc I_B = 1 Adc Pulsed; see 4.3.1			V_{CE}(sat)	---	0.9	Vdc
3071	Collector-emitter saturation voltage	I_C = 20 Adc I_B = 2 Adc Pulsed; see 4.3.1			V_{CE}(sat)	---	1.8	Vdc
	Subgroup 4		10	5				
3306	Magnitude of common emitter small signal short circuit transfer ratio	I_C = 2 Adc V_{CE} = 15 Vdc f = 10 MHz			/hfe/	1.0	---	8
3251	Pulse response:	Test Cond. A, except test circuit and pulse requirements per Figure 3 herein I_C = 20A						
3251	Turn-on time Turn-off time	--- ---			t_{on} t_{off}	--- ---	1.1 1.2	usec usec
	Subgroup 5		10	5				
	High temperature operation	T_A = +150°C						
3041	Collector-emitter cutoff current	Bias Cond. C V_{CE} = 100 Vdc V_{BE} = 0			I_{CES}	---	5	mAdc
	Low temp. operation:	T_A = −65°C						
3076	Forward current transfer ratio	Pulsed; see 4.3.1 I_C = 10 Adc V_{CE} = 10 Vdc			h_{FE}	10	---	---

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Table II. - Group B inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		20	20	---	---	---	---
2066	Physical dimensions 1/							
	Subgroup 2		15	15				
2031	Soldering heat	One cycle			---	---	---	---
1051	Temp. cycling	Test Cond. C 10 cycles; time at temp. extremes: 15 minutes (min.)			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. B			---	---	---	---
1021	Moisture resistance	no initial conditioning						
	End Point Tests:							
3061	Emitter-base cutoff current	Bias Cond. D V_{EB} = 8 Vdc			I_{EBO}	---	100	uAdc
3041	Collector-emitter cutoff current	Bias Cond. C V_{CE} = 100 Vdc V_{BE} = 0			I_{CES}	---	100	uAdc
3076	Forward current transfer ratio	Pulsed; see 4.3.1 I_C = 10 Adc V_{CE} = 10 Vdc			h_{FE}	30	120	---
	Subgroup 3		15	15				
2016	Shock	Non-operating 1500G, 5 blows of 0.5 msec each in orientations X1, Y1, Y2, Z1 total 20 blows.						
2056	Vibration, variable frequency				---	---	---	---
2006	Constant acceleration (centrifugal)	5,000 G's Orientations X1, Y1, Y2, Z1			---	---	---	---
	End Point Tests: Same as for Subgroup 2 above							

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Table II. - Group B inspection (cont'd)

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 4		20	20				
2036	Terminal strength terminal torque *base & emitter* only 2N5958, all terminal 2N5960	Test Cond. D1 *Torque = 6 in-oz* t = 15 sec.						
2036	terminal strength *stud torque*	Test Cond. A *Torque = 20 in-lbs.* 2/			---	---	---	---
	Measurements during stud torque test:	End Point tests as listed under Subgroup 2			---	---	---	---
1071	Seal	Fine leak test: condition G or H; gross leak test: condition A, C, D, or F			---	---	$5x10^{-7}$	atm cc/sec
	Subgroup 5		20	20				
1041	salt atmosphere (corrosion)	3/			---	---	---	---
	Subgroup 6		10	10				
3051	Safe operating area (dc operation)	4/ $T_C = +100^oC$ t = 60 sec t_r = 6 sec t_f = 6 sec 1 cycle			---	---	---	---
	Test 1:	P_T = 100W I_C = 20 Adc V_{CE} = 5 Vdc						
	Test 2:	P_T = 100W I_C = 5 Adc V_{CE} = 20 Vdc						

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Table II. - Group B inspection. (cont'd)

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 6 (cont'd)							
	Test 3:	T_C = +150°C P_T = 25W V_{CE} = 100 Vdc I_C = 250 mAdc						
	End Point Tests:							
	Same as above for Subgroup 2							
	Subgroup 7		10	10				
3053	Unclamped inductive sweep	T_A = +25°C I_B = 0.4A I_C = 3A L = 5 MH (see 4.5) Load cond. C			---	---	---	---
	End Point tests:							
	same as for Subgroup 2 above							
	Subgroup 8		10	10	---	---	---	---
3053	Clamped inductive sweep	T_A = +25°C I_B = 2A I_C = 20A L = 5 MH (see 4.6) Load Cond. B						
	End Point Tests:							
	Same as for Subgroup 2 above							
	Subgroup 9 5/		10	λ=7				
1031, 1032	High temp. life(non-operating)	Tstg. = +200°C t = 340 hours (for non-TX only)						
	End Point Tests:							
3061	Emitter-base cutoff current	Bias Cond. D V_{EB} = 8 Vdc			I_{EBO}	---	200	uAdc
3041	Collector-emitter cutoff current	Bias Cond. C V_{CE} = 100 Vdc V_{BE} = 0			I_{CES}	---	6/	7/
3076	Forward current transfer ratio	Pulsed; see 4.3.1 I_C = 10 Adc V_{CE} = 10 Vdc			h_{FE}	---	8/	7/

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Table II – Group B Inspection (cont'd)

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
1026, 1027	Subgroup 10							
	Steady state operation life:	T_J = +200°C ± 12.5°C t = 340 hrs. (for non-TX only) V_{CE} = 10 to 25 volts 5/	10	λ=7	---	---	---	---
	End Point Tests:							
	Same as for Subgroup 9 above							

1/ See 4.3.3 herein.

2/ With the transistor mounted in normal mounting position, the specified torque shall be maintained (torque wrench removed) throughout each of the specified End Point Tests and measurements.

3/ Electrical rejects from the same lot under evaluation may be used for this test.

4/ See 4.3.4 herein.

5/ See 4.2.3, 4.3.6 and 4.3.7 herein.

6/ ±100% or 20 uA, whichever is greater.

7/ Change from initial Group A reading.

8/ ±20%; however, $h_{FE} \geq 30$ shall be met.

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Table III. - Group C inspection.

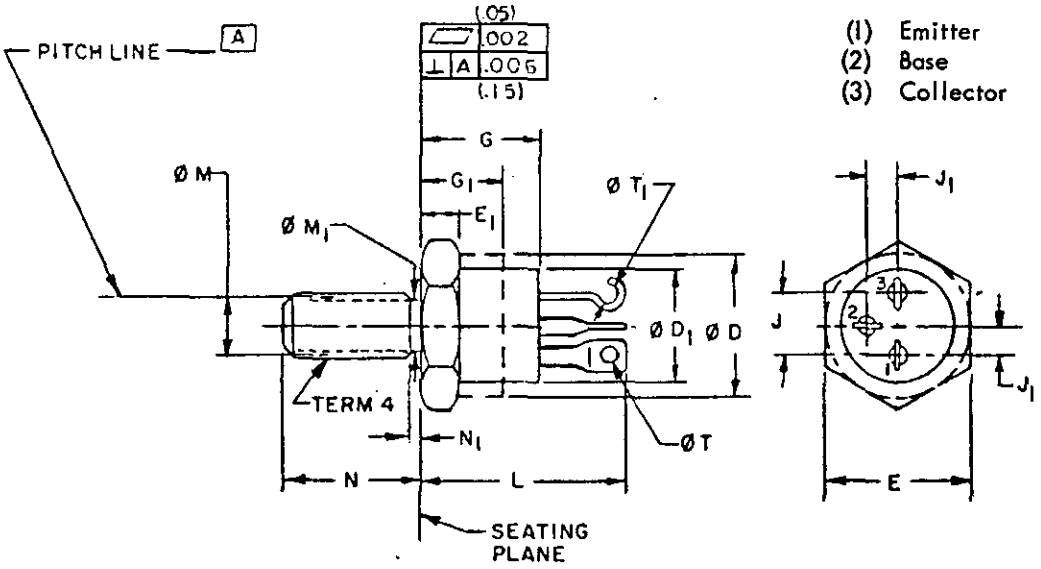
Test Method per MIL-STD-750	Examination or test	Conditions	LTPD Non-TX	LTPD TX	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		20	20				
3151	Thermal resistance (Junction to case)	I_C = 1 A V_{CE} = 10 V			θ_{J-C}	---	1.0	°C/W
1051	Temp. cycling	Test Cond. C1 except: t = 15 min at each temperature extreme			---	---	---	---
	End Point Tests:	*Same as Subgroup 2, Group B*						
	Subgroup 2		λ = 10					
1031	High temp. life (non-operating)	Tstg = +200°C t = 1000 hours			---	---	---	---
3061	Emitter-base cutoff current	Bias Cond. D V_{EB} = 8 Vdc			I_{EBO}	---	200	uAdc
3041	Collector-emitter cutoff current	Bias Cond. C V_{CE} = 100 Vdc V_{BE} = 0			I_{CES}	---	1/	2/
3076	Forward current transfer ratio	Pulsed; see 4.3.1 I_C = 10 Adc V_{CE} = 10 Vdc			h_{FE}	---	3/	2/
1026	Subgroup 3		λ = 10					
	Steady state operation life:	T_J = +200°C ± 12.5°C V_{CE} = 10 to 25 volts (see 4.2.3 and 4.3.6)		---	---	---	---	---
	End Point Tests:	Same as for Subgroup 2 above						
1027	Subgroup 4		10	7				
	Steady state operation life:	T_C = 100°C t = 340 hrs. V_{CE} = 10 to 25 volts P_T = 106W (see 4.3.7)		---	---	---	---	---
	End Point Tests:	same as for Subgroup 2 above						

1/ ±100% or 20 uA, whichever is greater.

2/ Change from initial Group A reading.

3/ ±20%; however, $h_{FE} \geq 30$ shall be met.

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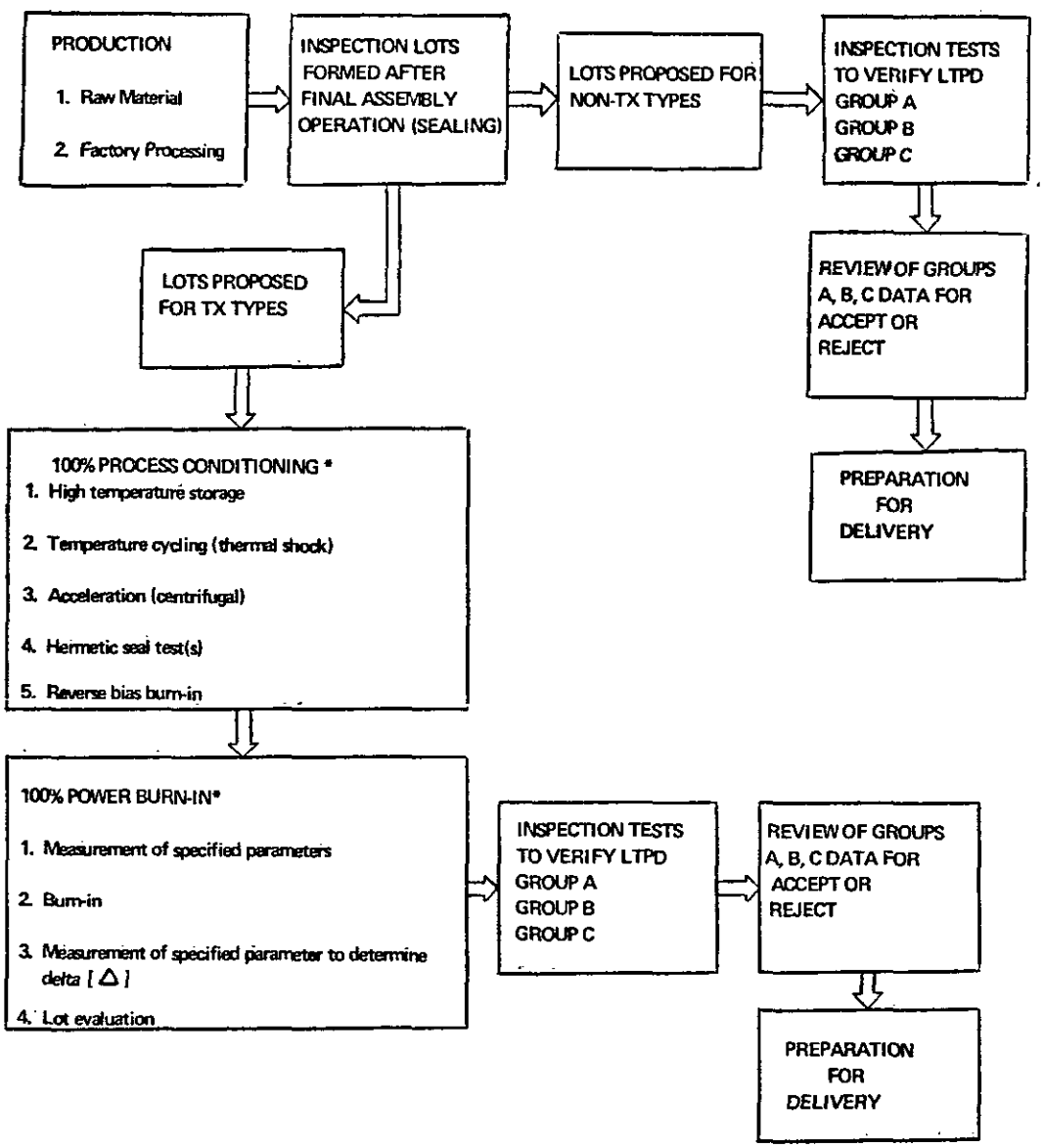


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ØD	.610	.687	15.50	17.44	2
ØD1	.570	.610	14.48	15.49	
E	.669	.688	17.00	17.47	
E1	.090	.150	2.29	3.81	1
G	.325	.460	8.26	11.68	
G1		.270		6.85	2
J1	.170	.213	4.32	5.41	5
J	.340	.415	8.64	10.54	5
L	.640	.875	16.26	22.22	
ØM					3
ØM1	.220	.249	5.59	6.32	
N	.422	.455	10.72	11.55	
N1		.090		2.28	6
ØT	.047	.072	1.20	1.82	
ØT1	.046	.077	1.17	1.95	4

NOTES
1. Chamfer or undercut on one or both ends of hexagonal portion is optional.
2. Package contour with the exception of the hexagon is optional within dimensions specified.
3. Pitch diameter 1/4-28 UNF-2A (coated) -. 2268 (5.76 mm).
4. This terminal can be flattened and pierced or hook type.
5. Position of leads in relation to the hexagon is not controlled.
6. Length of incomplete or undercut threads of ØM1.
7. 2N5958 Collector is electrically connected to case.
8. 2N5960 All leads electrically isolated from case.

FIGURE 1. Outline and dimension.

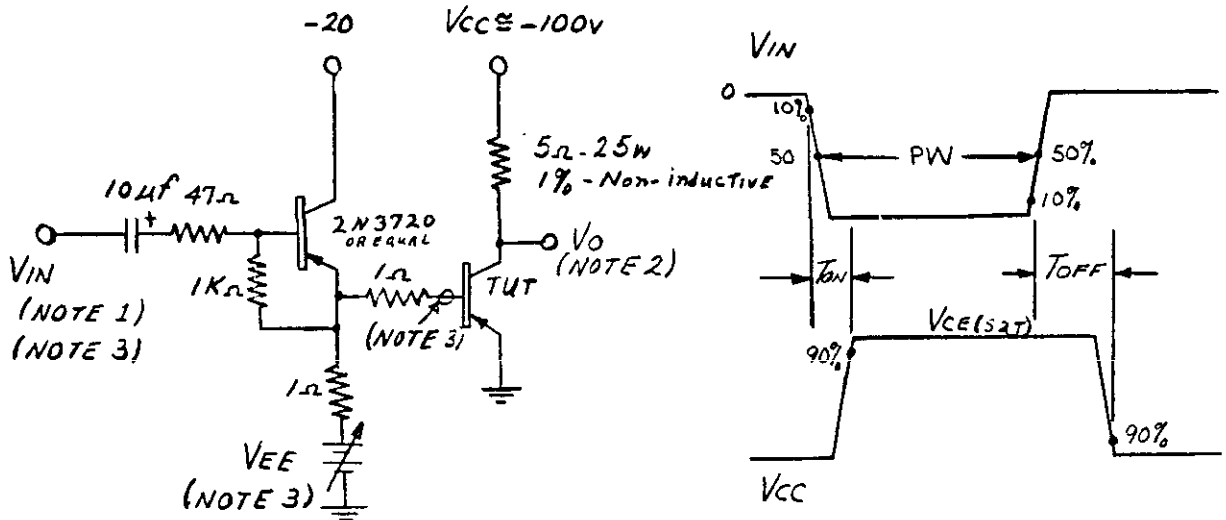
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* ORDER OF TESTS SHALL BE AS SHOWN WITHIN BLOCK.

Figure 2. Flow Chart of Non-TX and TX Types.

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NOTE 1 VIN (or equivalent)
GEN = HP214A
PW = 20 us
DC = 1%
Ampl. = see NOTE 3
tr = tf ≤15 ns

NOTE 2 VO (or equivalent)
1) Measured with a TEK. P6032 Probe.
2) Scope: TEK. 567
TEK. 3T77A
TEK. 6R1

NOTE 3 IB1 = IB2 = 10% IC
1) Current set with a TEK. P6042 probe.
2) Adjust Vin & VEE for correct base drive.

FIGURE 3 Switching time test circuit.

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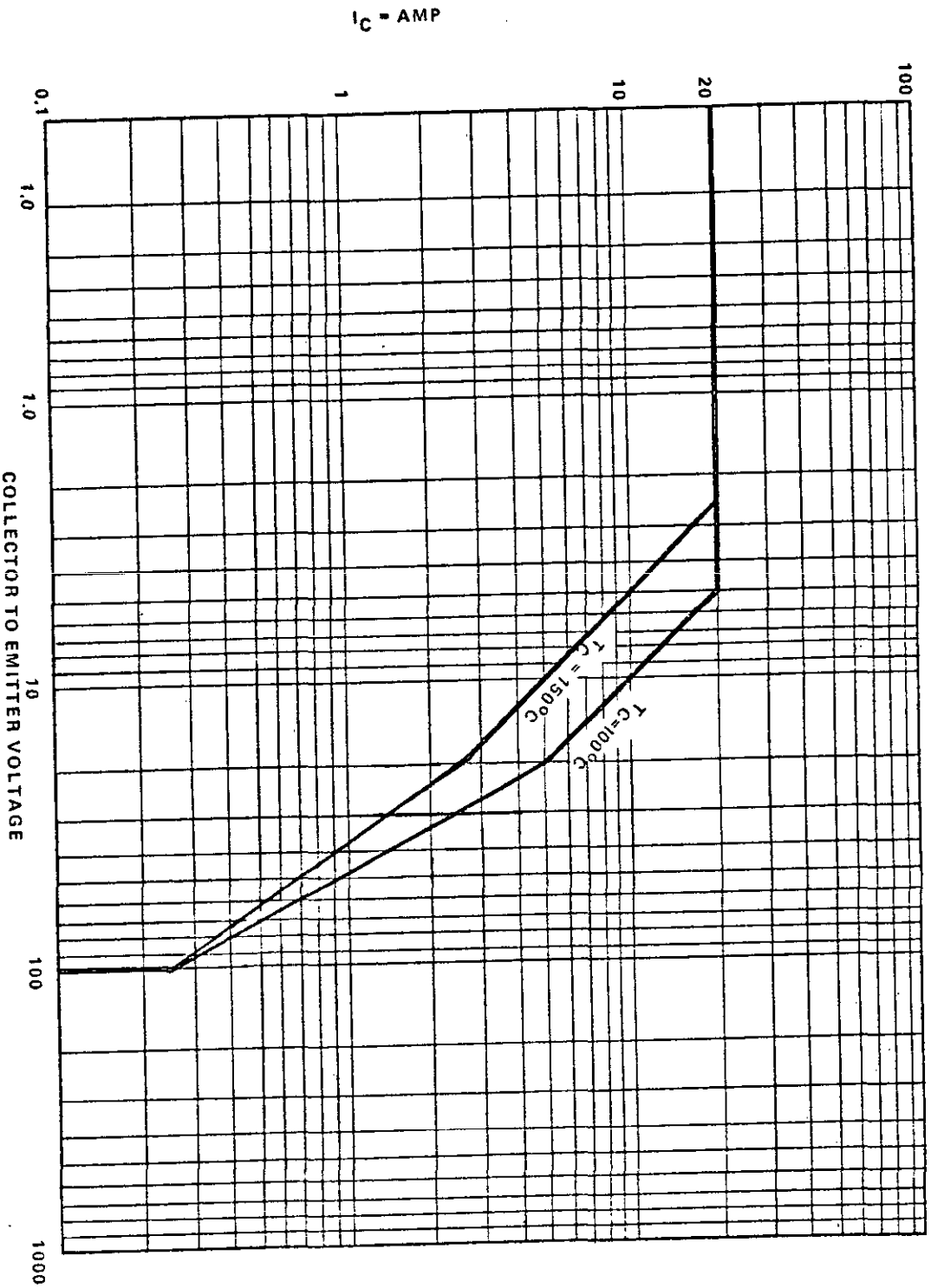


FIGURE 4. D.C. Safe Operating Area Nomograph.

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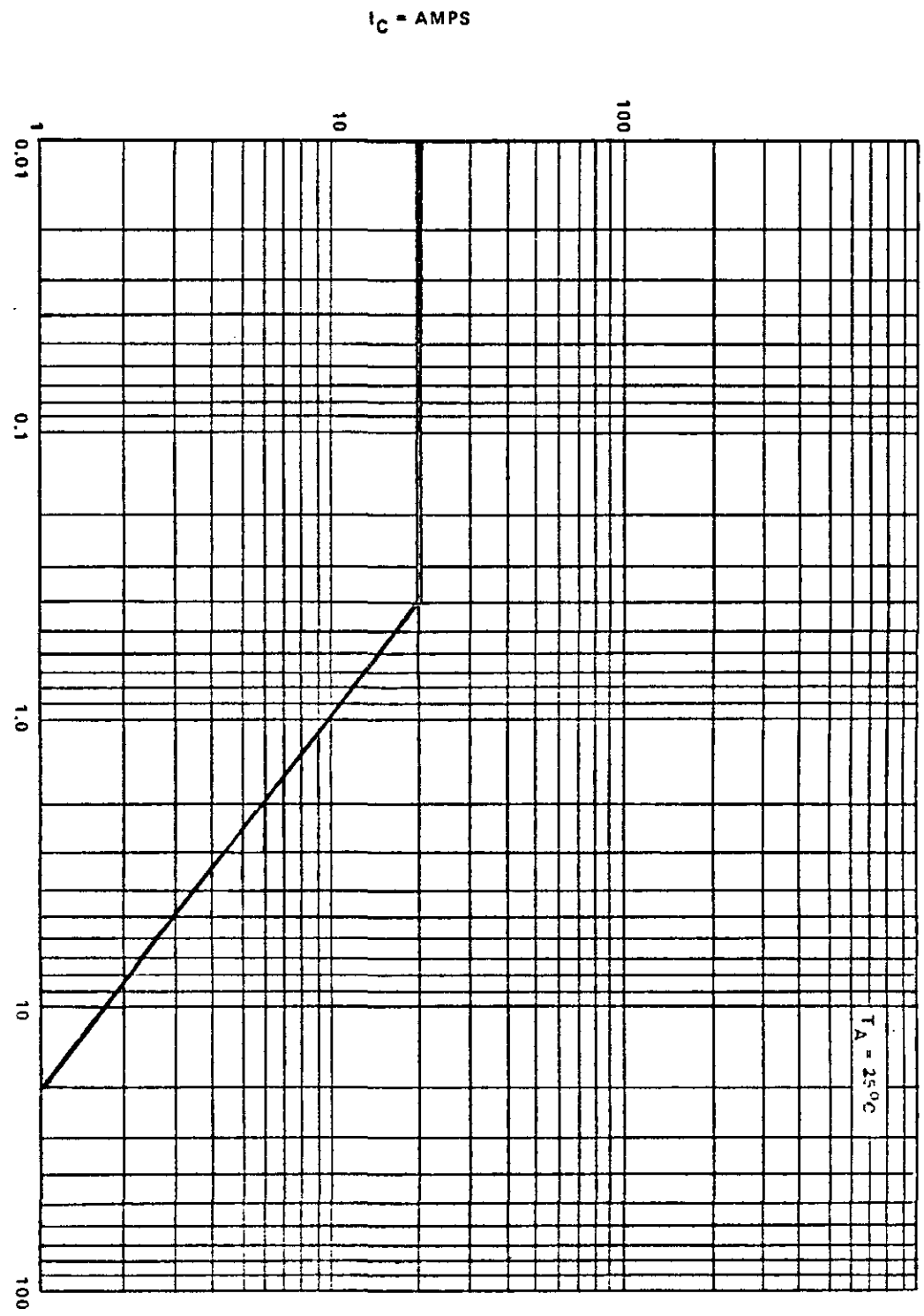


Figure 5. Switching between saturation and cutoff. Unclamped inductive load.

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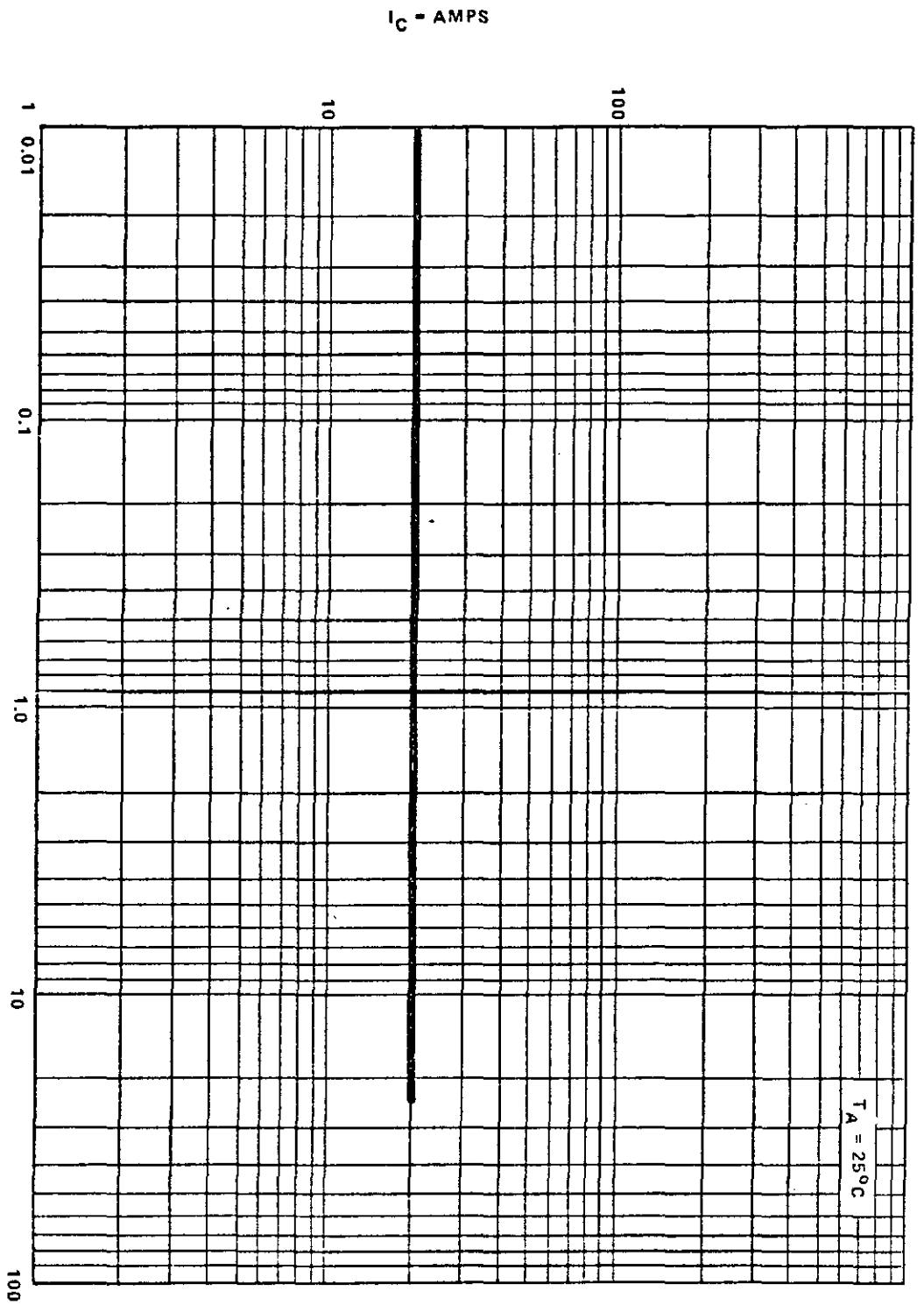


Figure 6. Safe Operating Area for switching between saturation and cutoff – clamped inductive load.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. - Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes. - The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance. - To insure proper circuit application, particular attention should be given to the differential voltage and gain characteristics, and ratings, pertinent to the individual transistor types covered herein.

6.3 Ordering data. - Lot inspection data: see 4.2.5 herein.

6.4 Procurement of "TX" types. - See 4.2.2 herein.

6.5 Quality assurance for "TX" types. - It is expected that where the permanent lot record, maintained as required in 4.4.1 herein, begins to show a condition of failures in excess of the LTPD rate specified herein, or a condition of entire-lot rejection (see 4.4.7.4 herein) is recorded, the Government procuring activity will determine any additional monitoring controls and evaluation criteria needed to attain restoration of the prescribed (herein) quality assurance for acceptance of product.

6.6 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U.S. Army Electronics Command, Fort Monmouth, New Jersey, Attention: AMSEL-PE-EM-2.

CUSTODIAN

ARMY-EL

Preparing Activity:

ARMY-EL

Project No. 5961 - A548

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